

		or the gravitational potential difference would also decrease.
12	A	Initial momentum of particle = mv Final momentum of particle = $-mv$ Change in momentum of particle = $-mv - (mv) = -2mv$ Time taken by particle between collisions with shaded side $t = \frac{2x}{v}$ Rate of change of momentum of particle = $\frac{-2mv}{\frac{2x}{v}} = \frac{-mv^2}{x}$ By N2L, the force exerted on the particle is $\frac{-mv^2}{x}$. The negative sign means that the force acting on the particle is pointing away from the shaded side. By N3L, the force on the shaded side of box = $-\left(\frac{-mv^2}{x}\right) = \frac{mv^2}{x}$
13	D	